TEAM ECLIPSE

AI Ambient Scribe

Mobile-First Clinical Documentation with Real-Time FHIR R4 Conversion — Built for Bharat's Healthcare 🇮🇳

PS-1 • Mobile-First Ambient AI Scribe with Real-Time FHIR Conversion

The Problem

CONTEXT

- Indian doctors spend 30-40% of time on manual documentation
- Most clinical conversations happen in Hinglish — no tools support it
- Paper prescriptions = no structured data for HMIS systems
- Tier 2/3 city clinics lack digital infrastructure

Objective: Capture doctor-patient conversations in real-time (Hindi + English mix), convert them into structured clinical notes, and map to **FHIR R4** compliant resources.

KEY STAKEHOLDERS

General Physicians

SME hospitals needing quick documentation

Junior Doctors & DMOs

High patient volume, minimal time per patient

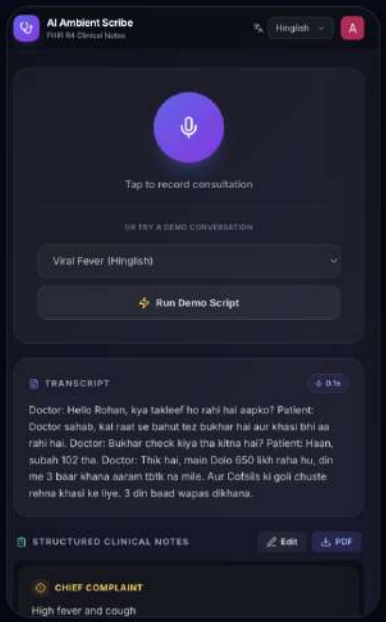
Clinic Owners — Tier 2/3 Cities

Need affordable digital documentation tools

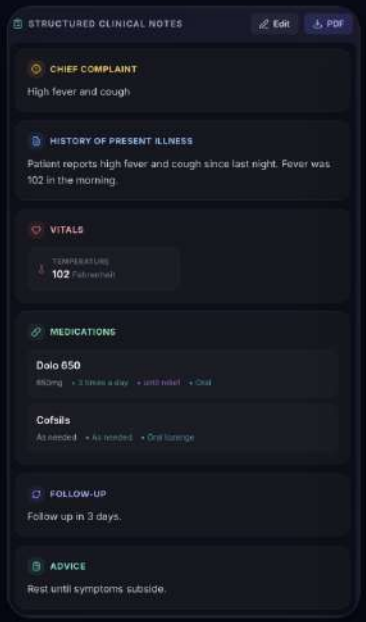
Live App Demo

Production-deployed mobile-first app — accessible from any phone browser

🔐 Secure Doctor Login



🎤 Recording & Hinglish Transcript



📋 Editable Clinical Notes + PDF

📦 FHIR R4 Bundle + Validation

Our Solution

A mobile-first AI-powered clinical scribe that listens, transcribes, structures, and validates — all in real-time.

Hinglish Voice → Text

Real-time transcription with **Doctor/Patient speaker tags** using Gemini 2.5 Flash. Supports Hindi, English & Hinglish.

Structured Clinical Notes

Auto-extracts Chief Complaint, HPI, Vitals, Diagnoses (ICD-10), Medications (RxNorm), Follow-up & Advice.

FHIR R4 Bundle

Generates Patient, Encounter, Observation, Condition & MedicationRequest with SNOMED-CT, LOINC coding.

Editable & Downloadable PDF

Doctors can edit AI-generated notes inline and download a polished prescription PDF instantly.

Clerk Authentication

Secure doctor sign-in with JWT-based route protection on both frontend and backend.

Deployed & Production-Ready

Frontend on **Vercel**, Backend on **Render** — accessible from any mobile browser.

System Architecture

📱 **Mobile Browser**
React 19 + Vite + TailwindCSS

→

🔐 **Clerk Auth**
JWT Token Verification

→

⚡ **FastAPI Backend**
Python + Uvicorn on Render

→

🧠 **Gemini 2.5 Flash**
Transcription + NLP

PIPELINE FLOW

🎤 Audio (WebM) → 📝 Hinglish Transcript → 📋 Structured Notes → 🏥 FHIR R4 Bundle → ✅ Validation

TECH STACK

FRONTEND	React 19 + TypeScript + Vite + Tailwind CSS	BACKEND	Python FastAPI + Uvicorn
AI ENGINE	Google Gemini 2.5 Flash	AUTH	Clerk (JWT + JWKS)
DATA STANDARD	HL7 FHIR R4	DEPLOYMENT	Vercel (FE) + Render (BE)

🎤 Hinglish Voice → Text with Speaker Tags

LIVE TRANSCRIPTION EXAMPLE

```
Doctor: Hello Rohan, kya takleef ho rahi hai aapko?
Patient: Doctor sahab, kal raat se bahut tez bukhar hai aur khasi bhi aa
rahi hai.
Doctor: Bukhar check kiya tha kitna hai?
Patient: Haan, subah 102 tha.
Doctor: Thik hai, main Dolo 650 likh raha hu...
```

3	Auto	<1s
Languages Supported	Speaker Diarization	Transcription Speed

KEY HIGHLIGHTS

- Gemini 2.5 Flash for audio-to-text
- Automatic Doctor/Patient speaker tagging
- Supports Hindi, English & Hinglish (code-mixed)
- Live waveform animation during recording
- Demo scripts for instant testing
- WebM audio capture via MediaRecorder API

📋 Structured Notes & FHIR R4 Bundle

AUTO-GENERATED CLINICAL NOTES

⚠️ Chief Complaint

High fever and cough

📝 HPI

Patient reports high fever and cough since last night. Fever was 102°F.

❤️ Vitals

Temperature: 102 Fahrenheit

💊 Medications

Dolo 650 (650mg, 3x/day) • Cofsils (as needed)

🔄 Follow-up

3 days

FHIR R4 RESOURCES GENERATED

Patient: 1 | Encounter: 1 | Condition: 2 | Observation: 1 | MedicationRequest: 2

✅ FHIR R4 Validation: PASSED — 7 resources • 0 warnings

CODING SYSTEMS USED

- **SNOMED CT** — Clinical findings & conditions
- **ICD-10** — Diagnosis codes
- **LOINC** — Observations & vitals
- **RxNorm** — Medications

📄 Editable Notes + Downloadable PDF

PS-1 Objective Mapping

OBJECTIVE	OUR IMPLEMENTATION	STATUS
Capture conversations in real-time (Hindi + English mix)	MediaRecorder API → Gemini 2.5 Flash transcription with Hinglish prompt & speaker diarization	✅ Done
Convert speech into structured clinical notes	Structured Notes: CC, HPI, Vitals, Diagnoses, Medications, Follow-up, Advice — all editable	✅ Done
Map entities to FHIR-compliant resources	Patient, Encounter, Observation, Condition, MedicationRequest with SNOMED/ICD-10/LOINC/RxNorm	✅ Done
Demonstrate documentation speed improvement	Real-time pipeline speed metrics — Transcription + FHIR processing time shown in app	✅ Done
Functional Prototype (Android/Mobile)	Mobile-first PWA-ready web app deployed on Vercel — works on any phone browser	✅ Done
FHIR Mapping Layer	Full FHIR R4 Bundle generation with auto-validation engine	✅ Done
Multilingual Capability	Hindi, English, and Hinglish support with language selector	✅ Done

🚀 Deployment & Security

PRODUCTION DEPLOYMENT

🌐 Frontend — Vercel

React 19 + Vite production build, global CDN, instant deployments, environment variable management for API URLs

⚙️ Backend — Render

FastAPI + Uvicorn with CORS configuration, auto-deploy from Git, environment secrets for Gemini API key

📡 API Endpoints

```
POST /api/transcribe/ — Audio → Text
POST /api/fhir/ — Transcript → FHIR + Notes
POST /api/fhir/validate — Bundle Validation
```

SECURITY & AUTHENTICATION

🔐 Clerk Authentication

JWT-based secure doctor sign-in. JWKS public key verification on backend. Protected API routes — no unauthenticated access.

🛡️ Backend Route Protection

Every API endpoint requires Bearer token. Token signature verified against Clerk's JWKS endpoint using RS256 algorithm.

🔒 Frontend Protection

SignedIn/SignedOut guards. Axios interceptor auto-attaches auth tokens. Unauthorized users see login-only screen.

TEAM ECLIPSE

Meet the Team

PS	AR	AY	MK
Parth Singla	**Aditya Raj**	**Aryan**	**Manish Kumar**
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Thank You! 🙏

Built with ❤️ for Indian Healthcare